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Semiconductor storage apparatus capable of efficiently performing screening test

Abstract

Provided is a semiconductor storage apparatus having a storage apparatus driving circuit which can efficiently perform screening tests for the semiconductor memory device. A semiconductor memory device includes a wright-voltage supply circuit, a wright-voltage switching circuit, a bit line discharge control circuit, a bit line discharge circuit, and a memory array. The wright-voltage supply circuit is connected to the memory array through the wright-voltage switching circuit. The bit line discharge control circuit is connected to the memory array through the bit line discharge circuit.

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U.S. PATENT DOCUMENTS

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FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H0629364	12/1993	JP	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefits of Japanese application no. 2022-136614, filed on Aug. 30, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The present invention relates to a semiconductor storage apparatus.

Related Art

(3) FLOTOX (Floating Gate Tunnel Oxide) type EEPROM semiconductor storage apparatuses may not be able to write or read correct data due to insulation defects between adjacent wirings of semiconductor elements. In order to detect the location of such insulation defects, the manufacturing process of the semiconductor storage apparatus includes a step of applying voltage stress between adjacent wirings of the semiconductor elements for inspection.

SUMMARY

(4) Japanese Unexamined Patent Publication No. H6-29364 discloses the principle of inspection by applying voltage stress between adjacent wirings in a semiconductor storage apparatus. However, a specific circuit for generating the signal to apply the voltage stress is not disclosed. The object of the present invention is to provide a semiconductor storage apparatus having a storage cell driving circuit which can efficiently perform screening tests of the storage apparatus.

- (5) In an aspect of the present invention, a semiconductor storage apparatus includes a wright-voltage supply circuit, a wright-voltage switching circuit, a bit line discharge control circuit, a bit line discharge circuit, and a memory array. The wright-voltage supply circuit is connected to the memory array through the wright-voltage switching circuit, and the bit line discharge control circuit is connected to the memory array through the bit line discharge circuit.
- (6) The semiconductor storage apparatus of the present invention can efficiently perform screening tests of the semiconductor storage apparatus by applying voltage stress between adjacent wirings for inspection.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram illustrating an example of a semiconductor storage apparatus according to an embodiment of the present invention.
- (2) FIG. 2 is a circuit diagram illustrating an example of a wright-voltage supply circuit according to the embodiment of the present invention.
- (3) FIG. 3 is a circuit diagram illustrating an example of a memory array circuit according to the embodiment of the present invention.
- (4) FIG. 4 is a circuit diagram illustrating an example of a bit line discharge circuit according to the embodiment of the present invention.
- (5) FIG. 5 is a circuit diagram illustrating an example of a memory array circuit according to the embodiment of the present invention.
- (6) FIG. 6 is a circuit diagram illustrating an example of a wright-voltage switching circuit according to the embodiment of the present invention.
- (7) FIG. 7 is a circuit diagram illustrating an example of a memory array according to the embodiment of the present invention.
- (8) FIG. 8 is a diagram illustrating an example of a bit line discharge circuit according to the embodiment of the present invention.

DESCRIPTION OF THE EMBODIMENTS

- (9) Referring to the drawings, an embodiment of the present invention will be described. FIG. 1 is a block diagram illustrating an example of a semiconductor storage apparatus according to the embodiment.
- (10) Referring to FIG. 1, the configuration of the semiconductor storage apparatus of the embodiment will be described. The semiconductor storage apparatus of the embodiment includes a data latch circuit **100**, a wright-voltage supply circuit **750**, a wright-voltage switching circuit **250**, a bit line discharge control circuit **850**, a bit line discharge circuit **450**, a row decoder **600**, a column decoder **500**, and a memory array **300**. The wright-voltage switching circuit **250**, the memory array **300**, the bit line discharge circuit **450**, and the column decoder **500** constitute a memory array circuit **2000**.
- (11) The connection of the semiconductor storage apparatus of the embodiment will be described. The data latch circuit **100** and the wright-voltage supply circuit **750** are connected to the memory array **300** through the wright-voltage switching circuit **250**, and the bit line discharge control circuit **850** and the column decoder **500** are connected to the memory array **300** through the bit line discharge circuit **450**.
- (12) Referring to FIG. 2, the configuration of the wright-voltage supply circuit **750** of the embodiment will be described. The wright-voltage supply circuit **750** includes seven input ports, three output ports, four two-input OR circuits **751-754** from the first two-input OR circuit to the fourth two-input OR circuit, and three level shift circuits **950-952** from the first level shift circuit to the third level shift circuit. The first to third level shift circuits **950** to **952** include a power port

VIN, a first input port IN1X, a second input port IN2, and an output port OUT. The seven input ports include a wright-voltage input port **701**, a first erase cycle control signal input port **702**, a second erase cycle control signal input port **703**, a first wright-cycle control signal input port **704**, a second wright-cycle control signal input port **705**, an odd sense line test signal input port **756**, and an even sense line test signal input port **757**. The three output ports are an even sense line wright-voltage output port **767**, an odd sense line wright-voltage output port **768**, and a bit line wright-voltage output port **769**.

(13) The connection of the wright-voltage supply circuit **750** in the embodiment will be described. The wright-voltage input port **701** is connected to the power port VIN of the first level shift circuit **950**, the power port VIN of the second level shift circuit **951**, and the power port VIN of the third level shift circuit **952**. The first erase cycle control signal input port **702** is connected to a first input port of the first two-input OR circuit **751** and a first input port of the third two-input OR circuit **753**. The second erase cycle control signal input port **703** is connected to a first input port of the second two-input OR circuit **752** and a first input port of the fourth two-input OR circuit **754**. The odd sense line test signal input port **756** is connected to a second input port of the first two-input OR circuit **751** and a second input port of the second two-input OR circuit **752**. The even sense line test signal input port **757** is connected to a second input port of the third two-input OR circuit **753** and a second input port of the fourth two-input OR circuit **754**.

(14) An output port of the first two-input OR circuit **751** is connected to the first input port IN1X of the first level shift circuit **950**. An output port of the second two-input OR circuit **752** is connected to the second input port IN2 of the first level shift circuit **950**. An output port of the third two-input OR circuit **753** is connected to the first input port IN1X of the second level shift circuit **951**. An output port of the fourth two-input OR circuit **754** is connected to a second input port IN2 of the second level shift circuit **951**. The first wright-cycle control signal input port **704** is connected to the first input port IN1X of the third level shift circuit **952**. The second wright-cycle control signal input port **705** is connected to a second input port IN2 of the third level shift circuit **952**. In this manner, the wright-voltage supply circuit **750** of the embodiment outputs two types of voltages, odd sense line wright-voltage and even sense line wright-voltage.

(15) An output port OUT of the first level shift circuit **950** is connected to the odd sense line wright-voltage output port **768**. An output port OUT of the second level shift circuit **951** is connected to the even sense line wright-voltage output port **767**. An output port OUT of the third level shift circuit **952** is connected to the bit line wright-voltage output port **769**.

(16) Referring to FIG. 3, the configuration of the first level shift circuit **950** of the embodiment will be described. The first level shift circuit **950** includes the power port VIN, the first input port IN1X, the second input port IN2, the output port OUT, first to fifth P-channel MOS transistors (hereinafter referred to as PMOS transistors) **901** to **905**, first to third N-channel MOS transistors (hereinafter referred to as NMOS transistors) **906** to **908**, a first inverter **909**, and a second inverter **910**.

(17) The connection of the first level shift circuit **950** of the embodiment will be described. The power port VIN is connected to a source port of the first PMOS transistor **901**, a source port of the third PMOS transistor **903**, and a source port of the fifth PMOS transistor **905**, as well as back gate ports of the first to fifth PMOS transistors **901** to **905**. The first input port IN1X is connected to a gate port of the second PMOS transistor **902** and a gate port of the first NMOS transistor **906** through the first inverter **909**, and further through the second inverter **910** to the a gate port of the fourth PMOS transistor **904** and a gate port of the second NMOS transistor **907**. The second input port IN2 is connected to the gate port of the third NMOS transistor **908**. The source port and back gate port of the first to third NMOS transistors **906** to **908** are connected to a GND port.

(18) A drain port of the first PMOS transistor **901** is connected to a source port of the second PMOS transistor **902**. A drain port of the third PMOS transistor **903** is connected to a source port of the fourth PMOS transistor **904**. A drain port of the second PMOS transistor **902** is connected to a gate port of the third PMOS transistor **903**, a gate port of the fifth PMOS transistor **905**, and a drain

port of the first NMOS transistor **906**. A drain port of the fourth PMOS transistor **904** is connected to a gate port of the first PMOS transistor **901** and a drain port of the second NMOS transistor **907**. A drain port of the fifth PMOS transistor **905** is connected to a drain port of the third NMOS transistor **908** and an output port OUT. The second level shift circuit **951** and the third level shift circuit **952** have the same configuration as the first level shift circuit **950**. Note that the level shift circuit may have a different circuit configuration than the level shift circuit described in the embodiment, as long as the level shift circuit has similar functions.

(19) Referring to FIG. **4**, the configuration of the bit line discharge control circuit **850** in the embodiment will be described. The bit line discharge control circuit **850** includes a first erase cycle control signal input port **702A**, an odd bit line test signal input port **851**, an even bit line test signal input port **852**, an even bit line discharge control signal output port **862**, an odd bit line discharge control signal output port **863**, a seventh inverter **853**, a fifth two-input OR circuit **856**, and a sixth two-input OR circuit **857**.

(20) The connection of the bit line discharge control circuit **850** in the embodiment will be described. The first erase cycle control signal input port **702A** is connected to a first input port of the fifth two-input OR circuit **856** and a first input port of the sixth two-input OR circuit **857** through the seventh inverter **853**. The odd bit line test signal input port **851** is connected to a second input port of the fifth two-input OR circuit **856**. The even bit line test signal input port **852** is connected to a second input port of the sixth two-input OR circuit **857**. An output port of the fifth two-input OR circuit **856** is connected to the odd bit line discharge control signal output port **863**. An output port of the sixth two-input OR circuit **857** is connected to the even bit line discharge control signal output port **862**. In this manner, the bit line discharge control circuit **850** in the embodiment outputs two types of signals, an odd bit line discharge control signal and an even bit line discharge control signal.

(21) Referring to FIGS. **5** to **8**, the configuration of the memory array circuit **2000** in the embodiment will be described. FIG. **5** is a circuit diagram illustrating the entire memory array circuit **2000**, the data latch circuit **100**, and the row decoder **600**. In the embodiment, an example will be described in which a region having a NMOS transistor **311** of the memory array **300** is a selected address region **310**. Referring to FIGS. **6** to **8**, each circuit constituting the memory array circuit **2000** will be described in detail.

(22) Referring to FIG. **6**, the configuration of the wright-voltage switching circuit **250** in the embodiment will be described. The wright-voltage switching circuit **250** of the embodiment includes a bit line wright-voltage input port **205**, an even sense line wright-voltage input port **257**, an odd sense line wright-voltage input port **258**, and fourth to seventh NMOS transistors **253** to **256**. Although there are many NMOS transistors other than the transistors illustrated in the drawing, only the transistors related to the memory elements of interest will be described here.

(23) The connection of the wright-voltage switching circuit **250** in the embodiment will be described. The bit line wright-voltage input port **205** is connected to the bit line wright-voltage output port **769** of the wright-voltage supply circuit **750** outside the wright-voltage switching circuit **250**, and is connected to a source port of the fourth NMOS transistor **253** and a source port of the fifth NMOS transistor **254** inside the wright-voltage switching circuit **250**. The even sense line wright-voltage input port **257** is connected to the even sense line wright-voltage output port **767** of the wright-voltage supply circuit **750** outside the wright-voltage switching circuit **250**, and is connected to a source port of the sixth NMOS transistor **255** inside the wright-voltage switching circuit **250**. The odd sense line wright-voltage input port **258** is connected to the odd sense line wright-voltage output port **768** of the wright-voltage supply circuit **750** outside the wright-voltage switching circuit **250**, and is connected to a source port of the seventh NMOS transistor **256** inside the wright-voltage switching circuit **250**.

(24) The sixth NMOS transistor **255** has a gate port connected to an even sense line data latch signal line **201** of the data latch circuit **100**, and a drain port connected to an even sense line **301** of

the selected address region **310** of the memory array **300**. The seventh NMOS transistor **256** has a gate port connected to an odd sense line data latch signal line **202** of the data latch circuit **100**, and a drain port connected to an odd sense line **302** of the selected address region **310** of the memory array **300**. The fourth NMOS transistor **253** has a gate port connected to an even bit line data latch signal line **203** of the data latch circuit **100**, and a drain port connected to an even bit line **303** of the selected address region **310** of the memory array **300**. The fifth NMOS transistor **254** has a gate port connected to an odd bit line data latch signal line **204** of the data latch circuit **100**, and a drain port connected to an odd bit line **304** of the selected address region **310** of the memory array **300**.

(25) Referring to FIG. 7, the configuration of the memory array **300** of the embodiment will be described. As described above, in the embodiment, the region having the NMOS transistor **311** of the memory array **300** is selected as the selected address region **310**. Here, the configuration related to the selected address region **310** will be described.

(26) The memory array **300** of the embodiment includes eighth to fifteenth NMOS transistors **311** to **314**, **321** to **324**, first to fourth floating gate MOS transistors (hereinafter referred to as FG-MOS transistors) **331**, **332**, **341**, **342**, the even sense line **301**, the odd sense line **302**, the even bit line **303**, the odd bit line **304**, a word **0** selection line **305**, and a word **1** selection line **306**. The even sense line **301** and the odd sense line **302** are arranged adjacent to each other in the memory array **300**. Similarly, the even bit line **303** and the odd bit line **304** are arranged adjacent to each other in the memory array **300**.

(27) The connection of the memory array **300** of the embodiment will be described. The even sense line **301** is connected to a drain port of the eighth NMOS transistor **311** and a drain port of the twelfth NMOS transistor **321**. The odd sense line **302** is connected to a drain port of the ninth NMOS transistor **312** and a drain port of the thirteenth NMOS transistor **322**. The even bit line **303** is connected to the drain ports of the tenth NMOS transistor **313** and the fourteenth NMOS transistor **323**. The odd bit line **304** is connected to a drain port of the eleventh NMOS transistor **314** and a drain port of the fifteenth NMOS transistor **324**.

(28) The word **0** selection line **305** is connected to a gate port of the eighth NMOS transistor **311**, a gate port of the ninth NMOS transistor **312**, a gate port of the tenth NMOS transistor **313**, and a gate port of the eleventh NMOS transistor **314**. The word **1** selection line **306** is connected to a gate port of the twelfth NMOS transistor **321**, a gate port of the thirteenth NMOS transistor **322**, the fourteenth NMOS transistor **323**, and a gate port of the fifteenth NMOS transistor **324**.

(29) A source port of the eighth NMOS transistor **311** is connected to a gate port of the first FG-MOS transistor **331** and a gate port of the second FG-MOS transistor **332**. A source port of the tenth NMOS transistor **313** is connected to the drain port of the first FG-MOS transistor **331**. A source port of the eleventh NMOS transistor **314** is connected to a drain port of the second FG-MOS transistor **332**. A source port of the twelfth NMOS transistor **321** is connected to a gate port of the third FG-MOS transistor **341** and a gate port of the fourth FG-MOS transistor **342**. A source port of the fourteenth NMOS transistor **323** is connected to the drain port of the third FG-MOS transistor **341**. A source port of the fifteenth NMOS transistor **324** is connected to a drain port of the fourth FG-MOS transistor **342**.

(30) Referring to FIG. 8, the configuration of the bit line discharge circuit **450** of the embodiment will be described. The bit line discharge circuit **450** of the embodiment includes an even bit line discharge control signal input port **452**, an odd bit line discharge control signal input port **453**, a sixteenth NMOS transistor **454**, and a seventeenth NMOS transistor **455**. Although there are many NMOS transistors other than those illustrated in the drawing, only the transistors related to the memory elements of interest will be described here.

(31) The connection of the bit line discharge circuit **450** of the embodiment will be described. The even bit line discharge control signal input port **452** is connected to the even bit line discharge control signal output port **862** of the bit line discharge control circuit **850** outside the bit line discharge circuit **450**, and is connected to a gate port of the sixteenth NMOS transistor **454** inside

the bit line discharge circuit **450**. The odd bit line discharge control signal input port **453** is connected to the odd bit line discharge control signal output port **863** of the bit line discharge control circuit **850** outside the bit line discharge circuit **450**, and is connected to a gate port of the seventeenth NMOS transistor **455** inside the bit line discharge circuit **450**.

(32) A drain port of the sixteenth NMOS transistor **454** is connected to the even bit line **303** of the memory array **300**. A drain port of the seventeenth NMOS transistor **455** is connected to the odd bit line **304** of the memory array **300**. A source port of the sixteenth NMOS transistor **454** and a source port of the seventeenth NMOS transistor **455** are connected to a GND port.

(33) Referring to FIG. **8**, the configuration of the column decoder **500** of the embodiment will be described. The column decoder **500** of the embodiment includes a column **0** selection signal input port **501**, a sense line bias signal input port **503**, and eighteenth to twenty-first NMOS transistors **511** to **514**. Although there are many NMOS transistors other than those illustrated in the drawing, only the transistors related to the memory elements of interest will be described here.

(34) The connections of the column decoder **500** of the embodiment will be described. The eighteenth NMOS transistor **511** has a gate port connected to a sense line bias signal input port **503**, and a source port connected to the even sense line **301** of the memory array **300**. The nineteenth NMOS transistor **512** has a gate port connected to the sense line bias signal input port **503**, and a source port connected to the odd sense line **302** of the memory array **300**. The twentieth NMOS transistor **513** has a gate port connected to the column **0** selection signal input port **501**, and a source port connected to the even bit line **303** of the memory array **300**. The gate port of the twenty-first NMOS transistor **514** is connected to the column **0** selection signal input port **501**, and the source port is connected to the odd bit line **304** of the memory array **300**.

(35) [Operation of Odd Sense Line Screening Test]

(36) The operation of the odd sense line screening test of the semiconductor storage apparatus of the embodiment will be described. The odd sense line screening test of the semiconductor storage apparatus of the embodiment is performed in an erase cycle. From the wright-voltage supply circuit **750**, the wright-voltage input port **701** receives a voltage V_{PP} ; the first erase cycle control signal input port **702**, the second erase cycle control signal input port **703**, and the even sense line test signal input port **757** receive low-level signals; and the first wright-cycle control signal input port **704**, the second wright-cycle control signal input port **705**, and the odd sense line test signal input port **756** receives high-level signals. The wright-voltage supply circuit **750** outputs the voltage V_{PP} from the even sense line wright-voltage output port **767**, and outputs $0V$ ($=GND$ voltage) from the odd sense line wright-voltage output port **768** and the bit line wright-voltage output port **769**. Here, the voltage V_{PP} is a high voltage required for performing erase and write operations on EEPROM.

(37) From the bit line discharge control circuit **850**, the first erase cycle control signal input port **702** receives a low-level voltage input, and regardless of the input voltages at the odd bit line test signal input port **851** and the even bit line test signal input port **852**, high-level voltages are output from the even bit line discharge control signal output port **862** and the odd bit line discharge control signal output port **863**. The bit line discharge circuit **450** receives signals from the bit line discharge control circuit **850**, and the sixteenth NMOS transistor **454** and the seventeenth NMOS transistor **455** are turned on.

(38) The data latch circuit **100** outputs the voltage V_{PP} to the even sense line data latch signal line **201** and the odd sense line data latch signal line **202**. The wright-voltage switching circuit **250** is supplied with the voltage V_{PP} from the bit line wright-voltage output port **769** of the wright-voltage supply circuit **750** at the bit line wright-voltage input port **205**, supplied with the voltage V_{PP} from the even sense line wright-voltage output port **767** of the wright-voltage supply circuit **750** at the even sense line wright-voltage input port **257**, and supplied with $0V$ from the odd sense line wright-voltage output port **768** of the wright-voltage supply circuit **750** at the odd sense line wright-voltage input port **258**. In the wright-voltage switching circuit **250**, the sixteenth NMOS transistor **255** and the seventh NMOS transistor **256** are turned on, the voltage V_{PP} is output to the

even sense line **301**, and 0V is output to the odd sense line **302**.

(39) Between adjacent even sense line **301** and odd sense line **302** of the memory array **300**, a voltage difference (voltage VPP–0V) is generated. The odd sense line screening test may generate voltage stress to detect defects caused by adjacent sense line failures in advance.

(40) [Operation of the Even Sense Line Screening Test]

(41) The operation of the even sense line screening test of the semiconductor storage apparatus according to the embodiment will be described. The even sense line screening test of the semiconductor storage apparatus of the embodiment is performed in an erase cycle. From the wright-voltage supply circuit **750**, the wright-voltage input port **701** receives the voltage VPP input; the first erase cycle control signal input port **702**, the second erase cycle control signal input port **703**, and the odd sense line test signal input port **756** receive low-level signals; and the first wright-cycle control signal input port **704**, the second wright-cycle control signal input port **705**, and the even sense line test signal input port **757** receive high-level signals are output. The wright-voltage supply circuit **750** outputs the voltage VPP from the odd sense line wright-voltage output port **768**, and outputs 0V from the even sense line wright-voltage output port **767** and the bit line wright-voltage output port **769**.

(42) From the bit line discharge control circuit **850**, the first erase cycle control signal input port **702** receives a low-level voltage input, and regardless of the input voltages at the odd bit line test signal input port **851** and the even bit line test signal input port **852**, high-level voltages are output from the even bit line discharge control signal output port **862** and the odd bit line discharge control signal output port **863**. The bit line discharge circuit **450** receives signals from the bit line discharge control circuit **850**, and the sixteenth NMOS transistor **454** and the seventeenth NMOS transistor **455** are turned on.

(43) The data latch circuit **100** outputs the voltage VPP to the even sense line data latch signal line **201** and the odd sense line data latch signal line **202**. The wright-voltage switching circuit **250** is supplied with 0V from the bit line wright-voltage output port **769** of the wright-voltage supply circuit **750** at the bit line wright-voltage input port **205**, supplied with 0V from the even sense line wright-voltage output port **767** of the wright-voltage supply circuit **750** at the even sense line wright-voltage input port **257**, and supplied with the voltage VPP from the odd sense line wright-voltage output port **768** of the wright-voltage supply circuit **750** at the odd sense line wright-voltage input port **258**. In the wright-voltage switching circuit **250**, the sixth NMOS transistor **255** and the seventh NMOS transistor **256** are turned on, 0V is output to the even sense line **301**, and the voltage VPP is output to the odd sense line **302**.

(44) Between the adjacent even sense line **301** and odd sense line **302** of the memory array **300**, a voltage difference (0V–voltage VPP) is generated. The even sense line screening test may generate voltage stress to detect defects caused by adjacent sense line failures in advance.

(45) [Operation of Odd Bit Line Screening Test]

(46) An operation of an odd bit line screening test of a semiconductor storage apparatus according to the embodiment will be described. The odd bit line screening test of the semiconductor storage apparatus of the embodiment is performed in a wright-cycle. From the wright-voltage supply circuit **750**, the wright-voltage input port **701** receives the voltage VPP; the first erase cycle control signal input port **702** and the second erase cycle control signal input port **703** receive high-level signals; and the first wright-cycle control signal input port **704**, the second wright-cycle control signal input port **705**, the odd sense line test signal input port **756**, and the even sense line test signal input port **757** receive low-level signals. The wright-voltage supply circuit **750** outputs 0V from the even sense line wright-voltage output port **767** and the odd sense line wright-voltage output port **768**, and outputs the voltage VPP from the bit line wright-voltage output port **769**.

(47) In the bit line discharge control circuit **850**, the first erase cycle control signal input port **702** and the odd bit line test signal input port **851** receive high-level voltages, and the even bit line test signal input port **852** receives a low-level voltage. The bit line discharge control circuit **850** outputs

a low-level voltage from the even bit line discharge control signal output port **862** and outputs a high-level voltage from the odd bit line discharge control signal output port **863**. The bit line discharge circuit **450** receives signals from the bit line discharge control circuit **850**, the sixteenth NMOS transistor **454** is turned off, and the seventeenth NMOS transistor **455** is turned on.

(48) The data latch circuit **100** outputs the voltage VPP to the even bit line data latch signal line **203** and the odd bit line data latch signal line **204** in the case of write data of the EEPROM being 0, and outputs 0V in the case of the write data of the EEPROM being 1. The wright-voltage switching circuit **250** is supplied with the voltage VPP from the bit line wright-voltage output port **769** of the wright-voltage supply circuit **750** at the bit line wright-voltage input port **205**, supplied with 0V from the even sense line wright-voltage output port **767** of the wright-voltage supply circuit **750** at the even sense line wright-voltage input port **257**, and supplied with 0V from the odd sense line wright-voltage output port **768** of the wright-voltage supply circuit **750** at the odd sense line wright-voltage input port **258**.

(49) The write data of the EEPROM for the even bit line data latch signal line **203** is set to 0, and the write data of the EEPROM for the odd bit line data latch signal line **204** is set to 1. In the wright-voltage switching circuit **250**, the fourth NMOS transistor **253** is turned on and the voltage VPP is output to the even bit line **303**, and the fifth NMOS transistor **254** is turned off and 0V is output to the odd bit line **304**.

(50) Between adjacent even bit line **303** and odd bit line **304** of the memory array **300**, a voltage difference (voltage VPP–0V) is generated. The odd bit line screening test may generate voltage stress for detecting defects caused by adjacent bit line failures in advance.

(51) By setting the write data of the EEPROM for the even bit line data latch signal line **203** to 1, and the write data of the EEPROM for the odd bit line data latch signal line **204** to 0, a similar potential difference (0V–voltage VPP) may be generated.

(52) [Normal Write Operation]

(53) The normal write operation of the semiconductor storage apparatus according to the embodiment will be described. In a wright-cycle, from the wright-voltage supply circuit **750**, the wright-voltage input port **701** receives the voltage VPP; the first erase cycle control signal input port **702** and the second erase cycle control signal input port **703** receive high-level signals; and the first wright-cycle control signal input port **704**, the second wright-cycle control signal input port **705**, the odd sense line test signal input port **756**, and the even sense line test signal input port **757** receive low-level signals. In the wright-cycle, the wright-voltage supply circuit **750** outputs 0V from the even sense line wright-voltage output port **767** and the odd sense line wright-voltage output port **768**, and outputs the voltage VPP from the bit line wright-voltage output port **769**.

(54) In the wright-cycle, from the bit line discharge control circuit **850**, the first erase cycle control signal input port **702** receives high-level signals, and the odd bit line test signal input port **851** and the even bit line test signal input port **852** receive low-level voltage inputs. In the wright-cycle, the bit line discharge control circuit **850** outputs low-level signals from the even bit line discharge control signal output port **862** and the odd bit line discharge control signal output port **863**. The bit line discharge circuit **450** receives signals from the bit line discharge control circuit **850**, and in the wright-cycle, the sixteenth NMOS transistor **454** and the seventeenth NMOS transistor **455** are turned off.

(55) The data latch circuit **100** outputs the voltage VPP to the even sense line data latch signal line **201** in the wright-cycle in the case of the even sense line data latch signal line **201** being selected as the address, and at the same time, the odd sense line data latch signal line **202** becomes a non-selected address, and 0V is output to the odd sense line data latch signal line **202** in the wright-cycle. In the case of the write data of even bits in EEPROM being 0 and the write data of odd bits being 1, the voltage VPP is output to the even bit line data latch signal line **203**, and 0V is output to the odd bit line data latch signal line **204**.

(56) In the wright-cycle, the wright-voltage switching circuit **250** is supplied with the voltage VPP

from the bit line wright-voltage output port **769** of the wright-voltage supply circuit **750** at the bit line wright-voltage input port **205**, supplied with 0V from the even sense line wright-voltage output port **767** of the wright-voltage supply circuit **750** at the even sense line wright-voltage input port **257**, and supplied with 0V from the odd sense line wright-voltage output port **768** of the wright-voltage supply circuit **750** at the odd sense line wright-voltage input port **258**. In the wright-voltage switching circuit **250**, the fourth NMOS transistor **253** and the sixth NMOS transistor **255** are turned on, the voltage VPP is output to the even bit line **303** and 0V to the even sense line **301**, the fifth NMOS transistor **254** and the seventh NMOS transistor **256** are turned off, and the odd bit line **304** and the odd sense line **302** are brought into a high impedance state.

(57) In the memory array **300**, the adjacent even sense line **301** and odd sense line **302** have one at 0V and the other in a high impedance state, while the adjacent even bit line **303** and the odd bit line **304** have one at the voltage VPP and the other in a high impedance state. There is no voltage applied between adjacent lines, and leakage of voltage between the lines can be suppressed. Note that in the case of the write data for even bits of EEPROM being 1 and the write data for odd bits being 0, the voltage states of the even bit line **303** and the odd bit line **304** are reversed, but similar effects can be obtained.

(58) [Normal Erase Operation]

(59) The normal erase operation of the semiconductor storage apparatus according to the embodiment will be described. In the erase cycle, from the wright-voltage supply circuit **750**, the wright-voltage input port **701** receives the voltage VPP; the first erase cycle control signal input port **702**, the second erase cycle control signal input port **703**, the odd sense line test signal input port **756**, and the even sense line test signal input port **757** receive low-level signals; and the first wright-cycle control signal input port **704** and the second wright-cycle control signal input port **705** receive high-level signals. In the erase cycle, the wright-voltage supply circuit **750** outputs the voltage VPP from the even sense line wright-voltage output port **767** and the odd sense line wright-voltage output port **768**, and outputs 0V from the bit line wright-voltage output port **769**.

(60) In the erase cycle, from the bit line discharge control circuit **850**, the first erase cycle control signal input port **702**, the odd bit line test signal input port **851**, and the even bit line test signal input port **852** receive low-level voltage inputs. In the erase cycle, the bit line discharge control circuit **850** outputs high-level signals from the even bit line discharge control signal output port **862** and the odd bit line discharge control signal output port **863**. The bit line discharge circuit **450** receives signals from the bit line discharge control circuit **850**, and in the erase cycle, the sixteenth NMOS transistor **454** and the seventeenth NMOS transistor **455** are turned on.

(61) In the erase cycle, the data latch circuit **100** outputs the voltage VPP to the even sense line data latch signal line **201** in the case of the even sense line data latch signal line **201** being selected as the address, and at the same time, the odd sense line data latch signal line **202** becomes a non-selected address, and 0V is output to the odd sense line data latch signal line **202**, and the voltage VPP is further output to the even bit line data latch signal line **203** and the odd bit line data latch signal line **204**.

(62) In the erase cycle, the wright-voltage switching circuit **250** is supplied with 0V from the bit line wright-voltage output port **769** of the wright-voltage supply circuit **750** at the bit line wright-voltage input port **205**, supplied with the voltage VPP from the even sense line wright-voltage output port **767** of the wright-voltage supply circuit **750** at the even sense line wright-voltage input port **257**, and supplied with the voltage VPP from the odd sense line wright-voltage output port **768** of the wright-voltage supply circuit **750** at the odd sense line wright-voltage input port **258**. In the wright-voltage switching circuit **250**, the fourth NMOS transistor **253**, a fifth NMOS transistor **254**, and the sixth NMOS transistor **255** are turned on, 0V is output to both the even bit line **303** and the odd bit line **304**, the voltage VPP is output to the even sense line **301**, the seventh NMOS transistor **256** is turned off, and the odd sense line **302** is brought into a high impedance state.

(63) In the memory array **300**, the adjacent even sense line **301** and odd sense line **302** have one at

the voltage VPP and the other in a high impedance state, while the adjacent even bit line **303** and odd bit line **304** are both at 0V. No voltage is applied between the adjacent lines, and the voltage leak between the lines can be suppressed.

(64) As described above, the semiconductor storage apparatus of the present invention can efficiently perform screening tests on semiconductor storage apparatuses by applying voltage stress between adjacent wires. Also, the semiconductor storage apparatus of the present invention may operate without applying voltage between adjacent wires during normal use, and can suppress leakage of the applied voltage. Although the semiconductor storage apparatus of the present invention has been described with examples of odd and even numbers, similar effects can be obtained with three or more types of signal examples.

Claims

1. A semiconductor storage apparatus, comprising: a wright-voltage switching circuit; a memory array; a wright-voltage supply circuit connected to the memory array through the wright-voltage switching circuit; a bit line discharge circuit; and a bit line discharge control circuit connected to the memory array through the bit line discharge circuit, wherein the wright-voltage supply circuit supplies at least two types of sense line wright-voltages to the wright-voltage switching circuit, wherein the wright-voltage switching circuit individually supplies voltages to at least two groups of sense lines in the memory array, wherein the bit line discharge control circuit outputs at least two types of bit line discharge control signals to the bit line discharge circuit, and wherein the bit line discharge circuit individually supplies discharge voltages to at least two groups of bit lines in the memory array.
2. The semiconductor storage apparatus according to claim 1, wherein the wright-voltage supply circuit comprises a wright-voltage input port, a first erase cycle control signal input port, a second erase cycle control signal input port, a first wright-cycle control signal input port, a second wright-cycle control signal input port, an odd sense line test signal input port, an even sense line test signal input port, an odd sense line wright-voltage output port, an even sense line wright-voltage output port, a bit line wright-voltage output port, first to fourth OR circuits, and first to third level shift circuits, wherein the wright-voltage input port is each connected to the first to third level shift circuits, the first OR circuit is connected to the first erase cycle control signal input port, the odd sense line test signal input port, and the first level shift circuit, the second OR circuit is connected to the second erase cycle control signal input port, the odd sense line test signal input port, and the first level shift circuit, the third OR circuit is connected to the first erase cycle control signal input port, the even sense line test signal input port, and the second level shift circuit, the fourth OR circuit is connected to the second erase cycle control signal input port, the even sense line test signal input port, and the second level shift circuit, the odd sense line wright-voltage output port is connected to the first level shift circuit, the even sense line wright-voltage output port is connected to the second level shift circuit, and the bit line wright-voltage output port is connected to the third level shift circuit.
3. The semiconductor storage apparatus according to claim 1, wherein the wright-voltage supply circuit comprises a wright-voltage input port, a first erase cycle control signal input port, a second erase cycle control signal input port, a first wright-cycle control signal input port, a second wright-cycle control signal input port, an odd sense line test signal input port, an even sense line test signal input port, an odd sense line wright-voltage output port, an even sense line wright-voltage output port, a bit line wright-voltage output port, first to fourth OR circuits, and first to third level shift circuits, wherein the wright-voltage input port is connected to the first to third level shift circuits, the first OR circuit is connected to the first erase cycle control signal input port, the odd sense line test signal input port, and the first level shift circuit, the second OR circuit is connected to the second erase cycle control signal input port, the odd sense line test signal input port, and the first

level shift circuit, the third OR circuit is connected to the first erase cycle control signal input port, the even sense line test signal input port, and the second level shift circuit, the fourth OR circuit is connected to the second erase cycle control signal input port, the even sense line test signal input port, and the second level shift circuit, the odd sense line wright-voltage output port is connected to the first level shift circuit; the even sense line wright-voltage output port is connected to the second level shift circuit, and the bit line wright-voltage output port is connected to the third level shift circuit.

4. The semiconductor storage apparatus according to claim 1, wherein the wright-voltage supply circuit comprises: first to seventh input ports; first to third output ports; a logic circuit including first to fourth input terminals connected to the first to fourth input ports, a first output terminal to which a logical addition result of a first signal supplied from the first input port and a second signal supplied from the second input port is supplied, a second output terminal to which a logical addition result of the second signal and a third signal supplied from the third input port is supplied, a third output terminal to which a logical addition result of the first signal and a fourth signal supplied from the fourth input port is supplied, and a fourth output terminal to which a logical addition result of the third signal and the fourth signal is supplied; a first level shift circuit including the first input terminal connected to the first output terminal of the logic circuit, the second input terminal connected to the second output terminal of the logic circuit, the third input terminal connected to the fifth input port, and an output terminal connected to the first output port; a second level shift circuit including the first input terminal connected to the third output terminal of the logic circuit, the second input terminal connected to the fourth output terminal of the logic circuit, the third input terminal connected to the fifth input port, and an output terminal connected to the second output port; and a third level shift circuit including the first input terminal connected to the sixth input port, the second input terminal connected to the seventh input port, the third input terminal connected to the fifth input port, and an output terminal connected to the third output port.

5. A semiconductor storage apparatus, comprising: a wright-voltage switching circuit; a memory array; a wright-voltage supply circuit connected to the memory array through the wright-voltage switching circuit; a bit line discharge circuit; and a bit line discharge control circuit connected to the memory array through the bit line discharge circuit, wherein the wright-voltage supply circuit comprises a wright-voltage input port, a first erase cycle control signal input port, a second erase cycle control signal input port, a first wright-cycle control signal input port, a second wright-cycle control signal input port, an odd sense line test signal input port, an even sense line test signal input port, an odd sense line wright-voltage output port, an even sense line wright-voltage output port, a bit line wright-voltage output port, first to fourth OR circuits, and first to third level shift circuits, wherein the wright-voltage input port is each connected to the first to third level shift circuits, the first OR circuit is connected to the first erase cycle control signal input port, the odd sense line test signal input port, and the first level shift circuit, the second OR circuit is connected to the second erase cycle control signal input port, the odd sense line test signal input port, and the first level shift circuit, the third OR circuit is connected to the first erase cycle control signal input port, the even sense line test signal input port, and the second level shift circuit, the fourth OR circuit is connected to the second erase cycle control signal input port, the even sense line test signal input port, and the second level shift circuit, the odd sense line wright-voltage output port is connected to the first level shift circuit, the even sense line wright-voltage output port is connected to the second level shift circuit, and the bit line wright-voltage output port is connected to the third level shift circuit.

6. A semiconductor storage apparatus, comprising: a wright-voltage switching circuit; a memory array; a wright-voltage supply circuit connected to the memory array through the wright-voltage switching circuit; a bit line discharge circuit; and a bit line discharge control circuit connected to the memory array through the bit line discharge circuit, wherein the wright-voltage supply circuit comprises: first to seventh input ports; first to third output ports; a logic circuit including first to fourth input terminals connected to the first to fourth input ports, a first output terminal to which a

logical addition result of a first signal supplied from the first input port and a second signal supplied from the second input port is supplied, a second output terminal to which a logical addition result of the second signal and a third signal supplied from the third input port is supplied, a third output terminal to which a logical addition result of the first signal and a fourth signal supplied from the fourth input port is supplied, and a fourth output terminal to which a logical addition result of the third signal and the fourth signal is supplied; a first level shift circuit including the first input terminal connected to the first output terminal of the logic circuit, the second input terminal connected to the second output terminal of the logic circuit, the third input terminal connected to the fifth input port, and an output terminal connected to the first output port; a second level shift circuit including the first input terminal connected to the third output terminal of the logic circuit, the second input terminal connected to the fourth output terminal of the logic circuit, the third input terminal connected to the fifth input port, and an output terminal connected to the second output port; and a third level shift circuit including the first input terminal connected to the sixth input port, the second input terminal connected to the seventh input port, the third input terminal connected to the fifth input port, and an output terminal connected to the third output port.
